Gas turbulence and convection
red supergiant convective envelope

Nikitina Maria

Foundation models for spatial-time series
Intelligent Systems Department, MIPT

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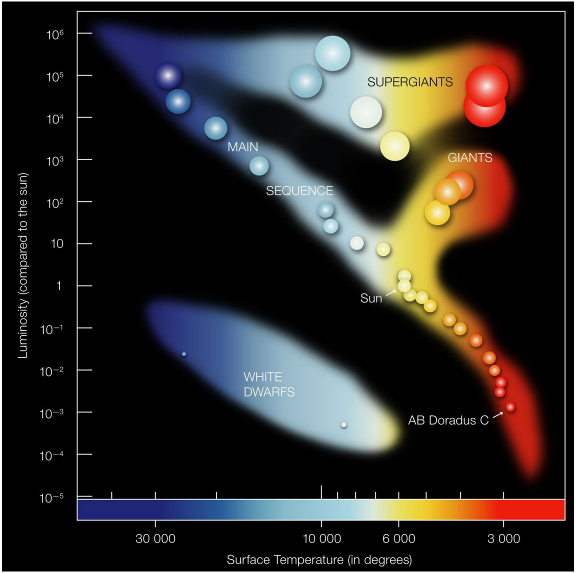
Problem Statement

As massive ($M \gtrsim 10M_\odot$) stars leave the main sequence, they expand to become red supergiants (RSGs), reaching radii of $\approx 300 - 1200R_\odot$ and luminosities of $\approx 10^4 - 10^{5.5}L_\odot$, approaching the **Eddington limit** and receiving increasing hydrostatic support from radiation pressure.

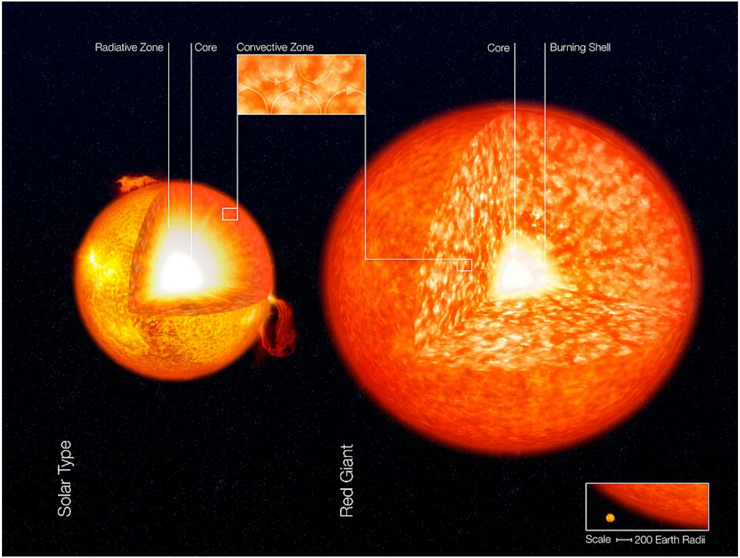
It is a theoretical challenge to realistically model stars, or even parts of stars, in 3D. This is especially true when radiative transfer must also be simultaneously solved through a highly turbulent medium with large density variations over optical depths. This is radiation hydrodynamic (RHD) challenge.

Jared A. Goldberg et.al. Numerical Simulations of Convective Three-dimensional Red Supergiant Envelopes, 2022

Hertzsprung–Russell diagram



Convection zone



1D Models

1D models represent a star as a set of concentric spherical layers, where all parameters depend only on the distance to the center. In 1D stellar evolution models, convection is conventionally handled by the **mixing-length theory (MLT)**.

The MLT approach derives from considering the fate of fluid elements as they move vertically, a distance referred to as the mixing length $l \equiv \alpha H$.

Definitions

Actual grad:

$$\nabla = \frac{d \ln T}{d \ln P}$$

Radiative grad:

$$\nabla_{rad} = -\frac{3\kappa\rho L_{surf} H}{64\pi r^2 \sigma_{SB} T^4}$$

Adiabatic grad:

$$\nabla_{ad} = \frac{d \ln T}{d \ln P}_{ad}$$

∇_e – grad inside an eddy as it moves.

Equations for MLP

Convection speed:

$$v_{conv}^2 = gQ(\nabla - \nabla_e)\frac{l^2}{\nu H}$$

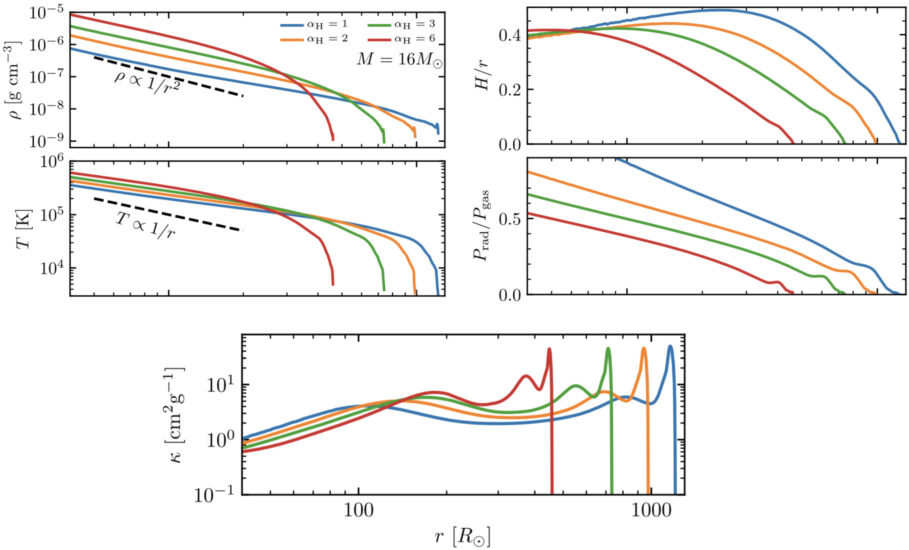
Convective flow:

$$F_{conv} = \rho c_P T \cdot v_{conv} \cdot (\nabla - \nabla_e) \cdot \frac{l}{H}$$

The condition of equality of flows:

$$F_{rad} + F_{conv} = \frac{L}{4\pi r^2} = const$$

1D experiment results



3D Model equations

Continuity Equation:

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{v}) = 0$$

Momentum Conservation:

$$\frac{\partial (\rho \boldsymbol{v})}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{v} \boldsymbol{v} + \mathrm{P}_{\mathrm{gas}}) = -\boldsymbol{G}_r - \rho \boldsymbol{\nabla} \Phi$$

Gas Energy Conservation:

$$\frac{\partial E}{\partial t} + \boldsymbol{\nabla} \cdot [(E + P_{\mathrm{gas}}) \boldsymbol{v}] = -c G_r^0 - \rho \boldsymbol{v} \cdot \boldsymbol{\nabla} \Phi,$$

where $E = E_g + \frac{1}{2} \rho v^2$.
Radiation Transport Equation:

$$\frac{\partial I}{\partial t} + c\, \boldsymbol{n} \cdot \boldsymbol{\nabla} I = S(I, \boldsymbol{n})$$

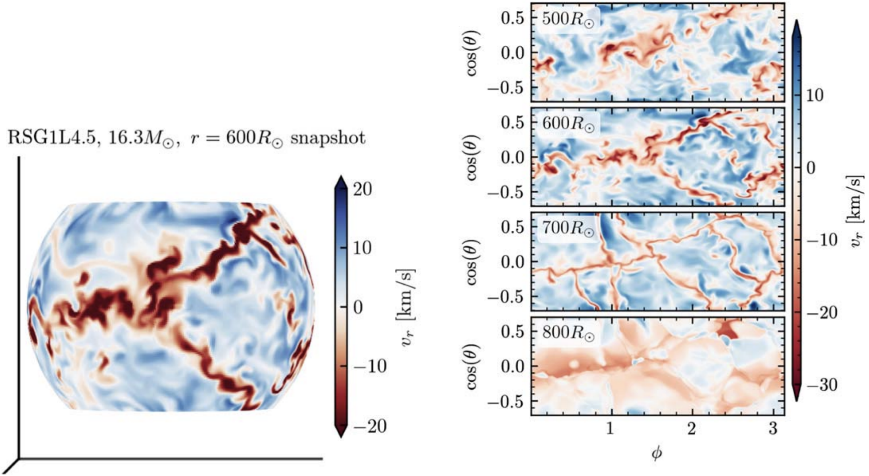
Modeling the interaction of matter and radiation

The source terms describing the interactions between gas and radiation in the comoving frame are:

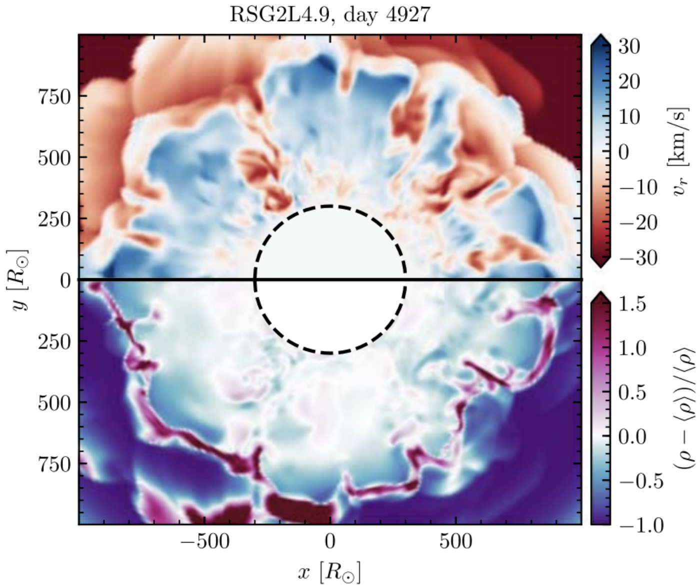
$$S_0(I_0, \boldsymbol{n}_0) = c\rho\kappa_{a\mathrm{P}} \left(\frac{ca_r T^4}{4\pi} - J_0 \right) + c\rho(\kappa_s + \kappa_{a\mathrm{R}})(J_0 - I_0),$$

$J_0 = \frac{1}{4\pi} \int I_0(\boldsymbol{n}_0) d\Omega_0$ is the mean intensity in the comoving frame.

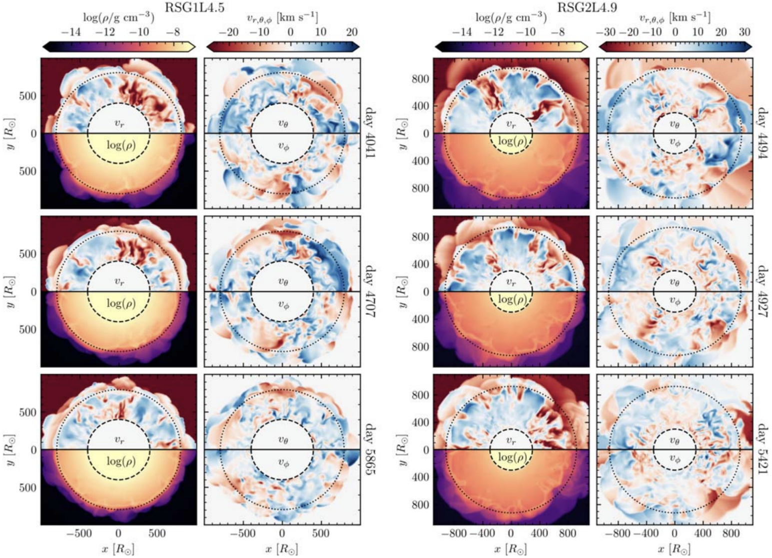
Results: radial velocity



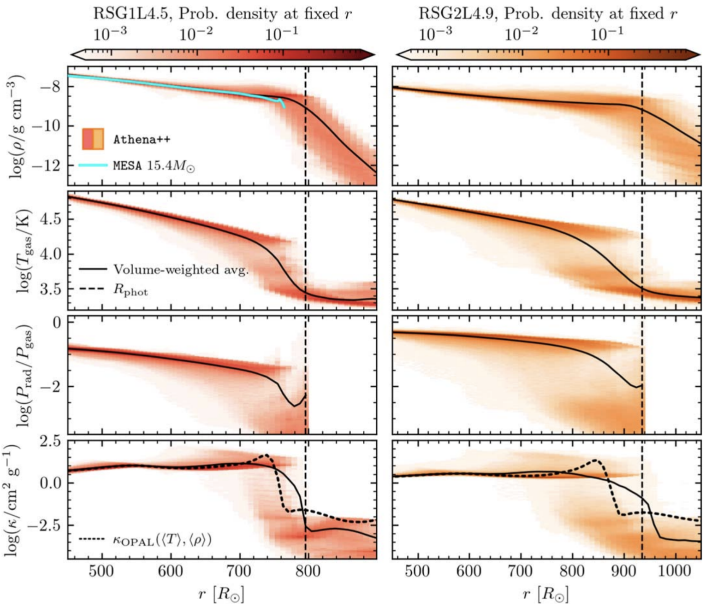
Results: radial velocity and density fluctuations



Results: radial velocity and density fluctuations



Comparing with 1D



Models results

Dataset	FNO	TFNO	Unet	CNextU-net
`convective_envelope_rsg`	**0.0269**	0.0283	0.0555	0.0799

Voxel Root Mean Square Error:

$$\text{VRMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(V_{\text{pred}}^{(i)} - V_{\text{true}}^{(i)})^2}$$

Conclusion

1. **Turbulence and convection**: These are inherently 3D processes that are difficult to model.
2. **Complex geometry**: Data is represented in spherical coordinates, and the volume of cells varies with the radius.
3. **The goal**: To create a fast "surrogate"simulator that could predict the evolution of the system without expensive direct numerical calculations based on the initial equations of hydrodynamics (which took 10 million CPU-hours).